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RESEARCH ARTICLE

Agentic AI With Orchestrator-Agent Trust: A Modular Visual Classification Framework With Trust-Aware Orchestration and RAG-Based Reasoning

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ABSTRACT Modern Artificial Intelligence (AI) increasingly relies on multi-agent architectures that blend visual and language understanding. Yet, a pressing challenge remains: How can we trust these agents especially in zero-shot settings with no fine-tuning? We introduce a novel modular Agentic AI visual classification framework that integrates generalist multimodal agents with a non-visual reasoning orchestrator and a Retrieval-Augmented Generation (RAG) module. Applied to apple leaf disease diagnosis, we benchmark three configurations: (I) zero-shot with confidence-based orchestration, (II) fine-tuned agents with improved performance, and (III) trust-calibrated orchestration enhanced by CLIP-based image retrieval and re-evaluation loops. Using confidence calibration metrics (ECE, OCR, CCC), the orchestrator modulates trust across agents. Our results demonstrate a 77.94% accuracy improvement in the zero-shot setting using trust-aware orchestration and RAG, achieving 85.63% overall. GPT-4o showed better calibration, while Qwen-2.5-VL displayed overconfidence. Furthermore, image-RAG grounded predictions with visually similar cases, enabling correction of agent overconfidence via iterative re-evaluation. The proposed system separates perception (vision agents) from meta-reasoning (orchestrator), enabling scalable and interpretable multi-agent AI. This blueprint illustrates how Agentic AI can deliver trustworthy, modular, and transparent reasoning, and is extensible to diagnostics, biology, and other trust-critical domains. In doing so, we highlight Agentic AI not just as an architecture but as a paradigm for building reliable multi-agent intelligence. All models, prompts, results, and system components including the complete software source code are openly released to support reproducibility, transparency, and community benchmarking at our Github page.

INDEX TERMS Agentic AI, orchestrator agent trust, trust orchestration, visual classification, retrieval augmented reasoning.

I. INTRODUCTION

The integration of vision and language models into autonomous decision-making systems has redefined the boundaries of artificial intelligence (AI), especially in fields

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that demand both perceptual accuracy and interpretability [1], [2]. Multimodal large language models (LMMs), capable of reasoning over both visual and textual data, are increasingly employed in diverse domains, ranging from autonomous robotics and medical imaging to scientific diagnostics and agricultural monitoring [3], [4], [5]. However, as these models are deployed in high-stakes environments, a fundamental challenge persists: can these systems be trusted to make reliable, transparent, and justifiable decisions, particularly in zero-shot or open-world scenarios where no prior task-specific fine-tuning is possible?

Recent developments in Agentic AI systems where autonomous agents collaborate, reason, and interact with their environment have highlighted the importance of meta-reasoning and modularity in AI architectures [6], [7], [8]. Rather than relying on monolithic end-to-end networks, Agentic AI systems distribute cognitive tasks across specialized AI agents [9]. In vision-language applications, this paradigm enables distinct agents to independently process visual inputs, generate explanations, and assess confidence, while an external orchestrator or supervisor performs higher-order reasoning to synthesize their outputs [10], [11], [12]. This architectural shift not only mirrors aspects of human collaborative problem-solving but also introduces an additional layer of oversight, which is critical for ensuring accountability and trust [13].

Despite the promise of this approach, trust calibration within Agentic AI remains an underexplored area. Conventional ensemble systems often assume that agents' self-reported confidence scores are reliable proxies for correctness [14], [15], [16], [17], an assumption that breaks down in zero-shot generalist models such as GPT-4o or Qwen-2.5-VL. These models may exhibit systematic overconfidence or fail to discriminate between subtle categories in domain-specific tasks [18]. Consequently, there is a pressing need to assess and correct misalignments between an agent's confidence and its actual performance. This problem is particularly acute in scientific and diagnostic domains, where misclassification may lead to erroneous conclusions or interventions.

In visual classification tasks such as plant disease detection, trustworthiness is not merely a matter of predictive accuracy but of interpretability and justification [19], [20], [21]. For instance, two models may arrive at the same prediction but differ significantly in the rationale behind their decision [22], [23]. A reliable AI system should not only be accurate but also capable of articulating why a decision was made and when to defer to alternative evidence [24]. RAG approaches, where models consult external databases or prior examples to refine their outputs, offer a promising pathway to ground predictions in visual context, enhancing both trust and transparency [25], [26], [27].

Deploying generalist VLMs in high-stakes, domain-specific scenarios presents critical challenges in calibration, interpretability, and decision trustworthiness. In agricultural automation and digital diagnostics, for instance, misclassifications can lead to delayed interventions and economic losses. Traditional deep learning pipelines offer high accuracy but lack modular reasoning and trust introspection.

To address these limitations, we propose a novel Agentic AI framework for visual classification that integrates trust-aware orchestration with retrieval-augmented reasoning in a modular, interpretable, and zero-shot-capable architecture. Our system coordinates multiple multimodal agents GPT-4o and Qwen-2.5-VL with a non-visual orchestrator that synthesizes final predictions based on reported confidence, natural language justifications, and internal trust scores. When discrepancies or low-confidence predictions arise, the orchestrator triggers a re-evaluation loop using RAG supported by CLIP-based image retrieval, allowing the system to reflect and refine its decision using similar visual precedents. We introduce a three-stage pipeline (see Figure 1), incrementally augmenting the system's capabilities.

The overall flow diagram and key contributions of this study are illustrated in Figure 2, highlighting how our trust-aware Agentic AI framework enables scalable integration of vision-language agents, dynamic trust calibration, and retrieval-augmented reasoning. Each experimental stage builds on the previous to increase reasoning fidelity and trust interpretability:

- **Experiment I:** AI Agents operate in zero-shot mode; the orchestrator selects final predictions using reported confidence scores.
- **Experiment II:** AI Agents are fine-tuned on a curated apple leaf disease dataset using hyperparameter optimization (Bayesian search) to evaluate the effects of supervised domain adaptation.
- **Experiment III:** Image-RAG, trust-aware orchestration, and agent re-evaluation are integrated into a unified Agentic AI system. A trust evaluation layer quantifies agent reliability using metrics such as Expected Calibration Error (ECE), Overconfidence Ratio (OCR), and Consistency Gap (CG), enabling the orchestrator to make informed arbitration decisions. The same general-purpose multimodal agents introduced in Experiment I independently classify images with natural language rationales and confidence scores; when trust metrics signal unreliability, the orchestrator initiates a re-evaluation loop that supplies agents with prior context and Image-RAG retrieval results. Agents then either revise or reaffirm their predictions, and the orchestrator makes a final decision based on these updated responses.

We validate this framework on a biologically grounded task: the classification of apple leaf diseases, where fine-grained distinctions (e.g., between rust and scab) and explanation interpretability are critical. Our results show an absolute improvement of 37.50 percentage points in classification accuracy, increasing from 48.13% in the baseline to 85.63% in the trust-aware configuration, even without

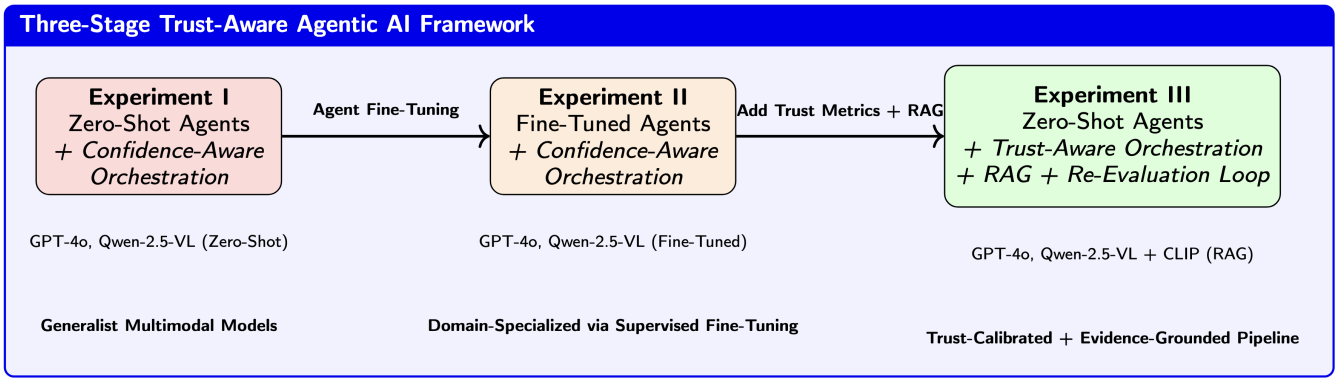

FIGURE 1. Three-stage trust-aware Agentic AI framework across experiments.

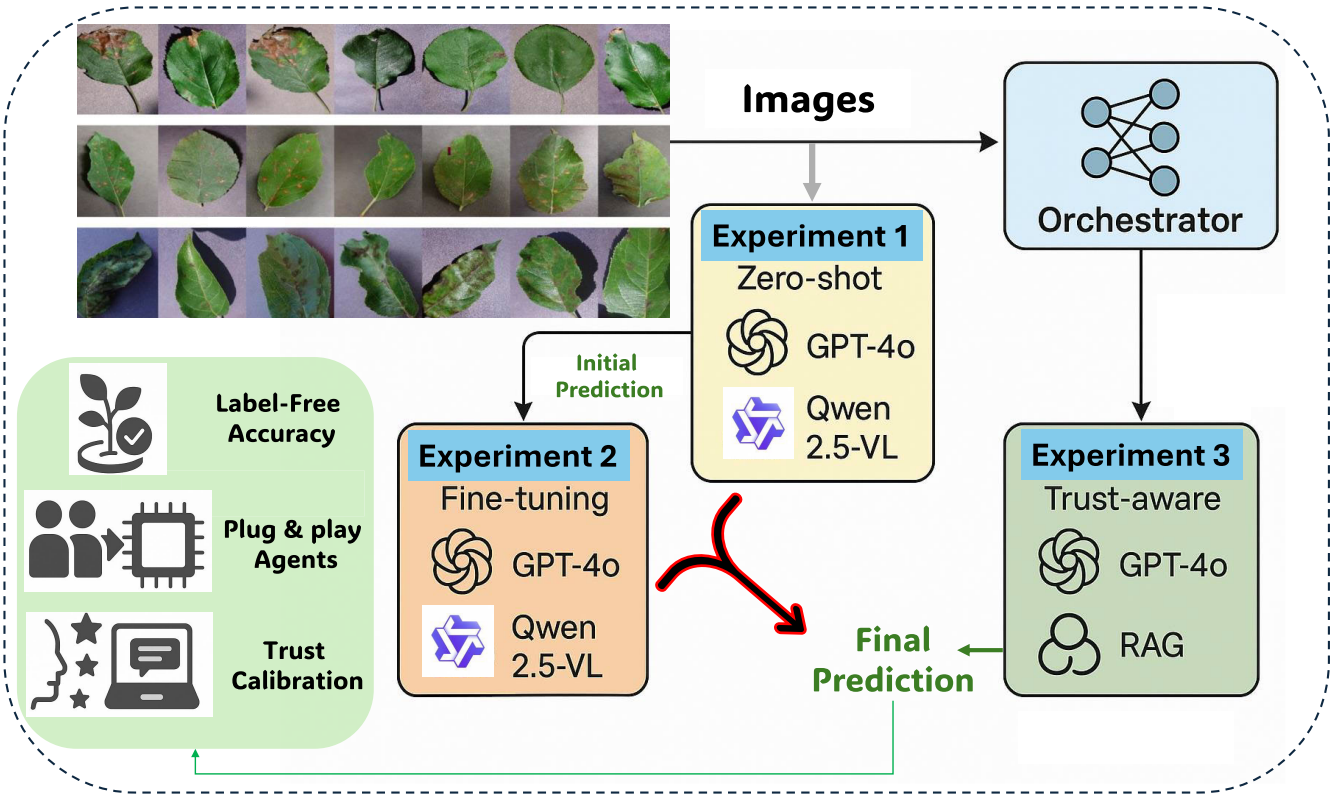

FIGURE 2. Overview of our trust-aware Agentic AI framework for visual classification, illustrating modular agents, orchestration stages, trust calibration, and retrieval-augmented re-evaluation for accurate, interpretable decisions. This workflow is designed for plant leaf disease classification but is generalizable to any RGB image classification task involving multimodal agents and trust-aware decision pipelines.

additional fine-tuning in Experiment III. This substantial improvement demonstrates the power of structured trust arbitration and retrieval-enhanced reasoning for improving performance and interpretability in real-world, open-domain applications.

In summary, our contributions are fourfold: (1) A modular Agentic AI system that decouples perception, reasoning, and retrieval; (2) a novel trust-aware orchestration strategy grounded in multi-dimensional calibration; (3) a CLIP-RAG-based re-evaluation loop for uncertainty mitigation; and (4) a comprehensive empirical validation across three reasoning regimes with reproducible and scalable design.

II. METHODOLOGY

This study proposes and evaluates a modular Agentic AI architecture for visual classification, combining two vision-language agents (GPT-4o and Qwen-2.5-VL) with a non-visual orchestrator (o3-mini-2025-01-31). Experiments were conducted using a curated dataset of 800 RGB images labeled into four apple disease categories, split

64% for training, 16% for validation, and 20% for testing.

The system comprises three experimental setups.

A. EXPERIMENT I

In the first experimental configuration, we evaluate the performance of general-purpose multimodal LLMs in a zero-shot setting that is, without any task-specific fine-tuning.

Each agent receives a prompt containing a single input image (from the test set) and is tasked with predicting the most appropriate plant disease class. In addition to the predicted class label, agents are required to return (i) a confidence score in the range [0.0–1.0] and (ii) a natural language explanation justifying their decision (Appendix, Fig. 9). These three outputs - classification, confidence, and rationale - are compiled and forwarded to a non-visual orchestrator model (o3-mini-2025-01-31 [28]), which serves as a comparative reasoner. The orchestrator does not process images directly; instead, it evaluates the agents' predictions and justifications in light of their associated confidence scores and produces a final classification decision through structured, confidence-aware reasoning (Appendix, Fig. 10).

B. EXPERIMENT II

In Experiment II, both agents were fine-tuned using supervised learning techniques to improve classification performance. For GPT-4o, fine-tuning employed a hyperparameter configuration informed by prior ResNet-50 optimization studies. Qwen-2.5-VL underwent over 50 hyperparameter tuning trials, beginning with heuristic parameter estimates and refined through performance-based search strategies.

1) **GPT-4o Fine-Tuning:** In this phase, we explored two fine-tuning strategies: one using default hyperparameters provided by the OpenAI platform, and one informed by prior hyperparameter optimization conducted on a ResNet-50 model using Bayesian optimization.

   a) ResNet-50–Informed Hyperparameter Transfer. To mitigate the computational cost of performing hyperparameter optimization directly on GPT-4o, we hypothesized that high-performing hyperparameters derived from ResNet-50 tuning on the same dataset could be effectively transferred to GPT-4o. Specifically, Bayesian optimization with the Tree-structured Parzen Estimator (TPE) algorithm was used to explore the ResNet-50 hyperparameter space across 30 trials. TPE iteratively models the objective function $f(x)$, evaluates the expected improvement (EI) of candidate configurations, and selects promising trials using a likelihood ratio.
   Optimization was implemented using the Optuna library on an NVIDIA A100-SXM4-40GB GPU via Colab Enterprise. Early stopping and pruning were used to improve computational efficiency. The best configuration identified 10 training epochs and a batch size of 16 was subsequently applied to GPT-4o fine-tuning through the OpenAI fine-tuning interface. Training and validation sets were formatted into `jsonl` files with prompt–completion pairs before submission.

   b) Default Hyperparameter Configuration. In parallel, we conducted fine-tuning using the default hyperparameters recommended by the OpenAI platform: 3 epochs and a batch size of 1. Identical training and validation files were used to ensure a fair comparison.

   Fine-tuning GPT-4o using the ResNet-50–informed hyperparameters (10 epochs, batch size 16) required approximately 1,778 seconds (∼29.6 minutes) and incurred a cost of USD 47.53. In comparison, fine-tuning with the default hyperparameters (3 epochs, batch size of 1) required 1,652 seconds (∼27.5 minutes) at a reduced cost of USD 13.09. Despite similar wall-clock times, the financial cost difference arises from the number of training tokens processed: the ResNet-50–informed configuration processes significantly more tokens due to higher epoch count and larger batch size. Although direct hyperparameter optimization on GPT-4o could potentially yield higher-performing configurations, the computational and financial cost of conducting such a process over multiple trials renders it impractical under current constraints.
   A detailed comparison of runtime, cost, and validation loss for both configurations is provided in the Appendix, Table 3.
   Based on these results, the ResNet-50–informed configuration yielded a substantially lower validation loss (0.0088) and was therefore selected as the preferred GPT-4o variant for integration into the Agentic AI system implemented in this study.

2) **Qwen-2.5-VL-7B Fine-Tuning:** To adapt the Qwen-2.5 Vision-Language 7B (VL-7B) model for our task, we fine-tuned the `unsloth/Qwen2.5-VL-7B-Instruct-bnb-4bit` model [29], a 4-bit quantized variant of the original pretrained Qwen2.5 model. This quantized version enables efficient loading and inference with reduced memory requirements, making it suitable for commodity GPU hardware. We employed Low-Rank Adaptation (LoRA) in 16-bit precision to inject trainable adapters into the model, thereby allowing effective fine-tuning without full dequantization of the backbone weights.
   The LoRA adapters were integrated into all major architectural components of the model, including the vision encoder layers, the language modeling transformer layers, multi-head attention modules, and the feedforward multilayer perceptrons (MLPs). Each adapter was configured with a rank $r = 16$ and scaling factor $\alpha = 16$, balancing parameter efficiency with

expressive capacity. We used the Parameter-Efficient Fine-Tuning (PEFT) framework to update only the LoRA-injected weights, keeping the underlying 4-bit quantized parameters frozen throughout training.

To identify the optimal hyperparameter configuration, we employed Bayesian optimization using the Optuna framework, leveraging the TPE as the sampler. The search space included learning rate, per-device batch size, gradient accumulation steps, warmup ratio, weight decay, and number of training epochs. Optimization targeted the minimization of validation loss. Trial results were persistently stored in both `jsonl` and SQLite formats, allowing for checkpointing and resumability in the event of interruption.

Poor-performing trials were automatically pruned using Optuna's `MedianPruner`, which compares intermediate results to the median of previous completed trials and terminates underperformers early. Early stopping was also employed during training to prevent overfitting and reduce unnecessary computational expenditure. All training was conducted using the `SFTTrainer` class from the Hugging Face `trl` library, which supports supervised fine-tuning with periodic evaluation at the end of each epoch. Throughout training, only the LoRA adapters were updated, while the quantized backbone remained frozen to preserve efficiency.

Recognizing the potential inefficiency of starting hyperparameter optimization from purely random initial conditions, particularly in high-dimensional or sensitive parameter spaces, we manually enqueued a strong initial configuration to guide the search. This warm-start configuration used a learning rate of 2×10^{-4}, a per-device batch size of 2, 8 gradient accumulation steps, a warmup ratio of 0.05, a weight decay of 0.01, and 10 training epochs. This guided initialization enabled the optimizer to begin exploring in a region of the parameter space known to yield promising results.

A total of 50 hyperparameter optimization trials were performed. Among these, 20 trials completed successfully, while 30 were pruned based on intermediate performance. No trials failed due to runtime errors. The best-performing trial achieved a validation loss of 1.0×10^{-5}, using a learning rate of approximately 1.094×10^{-5}, a batch size of 4, 4 gradient accumulation steps, a warmup ratio of 0.0997, a weight decay of 0.00127, and 15 training epochs (Appendix, Table 7).

During inference, both the fine-tuned GPT-4o and Qwen agents received a single input image and generated three outputs: (i) a predicted disease category, (ii) a natural language explanation justifying the classification, and (iii) a normalized confidence score in the range $[0, 1]$. As in the previous configuration (Experiment I), a non-visual orchestrator model (`o3-mini-2025-01-31`) served as the final decision-maker. This orchestrator performed comparative reasoning by evaluating structured prompts that included each agent's predictions, explanations, and confidence scores. Based on this synthesis, it produced a consolidated classification accompanied by a rationale for its selection.

C. EXPERIMENT III

In the final experimental configuration, we integrate CLIP-based image retrieval (Image-RAG), trust-aware orchestration, and agent re-evaluation mechanisms into our Agentic AI system. The orchestrator incorporates similarity-based retrieval results and calibration metrics to determine whether to trust agent outputs or trigger re-evaluation loops. The system dynamically guides agents to revise their predictions when confidence or justification is misaligned with expected trust signals (Fig. 11).

1) SYSTEM ARCHITECTURE AND WORKFLOW

The system operates as a hierarchical multi-agent architecture in which the orchestrator functions as a meta-reasoning layer positioned above both vision-language agents (GPT-4o and Qwen-2.5-VL). The workflow proceeds as follows:

1) **Parallel Agent Inference:** Given an input image, both GPT-4o and Qwen-2.5-VL agents independently and simultaneously process the image in zero-shot mode. Each agent generates three outputs: (i) a predicted disease class, (ii) a confidence score in [0, 1], and (iii) a natural language justification for its prediction.
2) **Orchestrator Synthesis:** The orchestrator (o3-mini-2025-01-31) receives the outputs from *both* agents simultaneously. Importantly, the orchestrator does not process the image directly; instead, it performs comparative meta-reasoning by synthesizing the predictions, confidence scores, and explanations from both agents. Using pre-computed trust profiles (ECE, OCR, CG) for each agent, the orchestrator evaluates the reliability of each agent's output.
3) **Trust-Based Decision:** If both agents agree and exhibit high trust scores, the orchestrator accepts the consensus prediction. If agents disagree or if trust metrics indicate unreliability (e.g., high ECE, high OCR, or low CCC), the orchestrator triggers a re-evaluation loop.
4) **Re-Evaluation Loop:** When triggered, the orchestrator queries the Image-RAG system to retrieve the top-k visually similar reference images along with their labels and confidence scores. This retrieval context is then provided back to both agents via an augmented prompt that includes: (a) their prior prediction, (b) the Image-RAG retrieval results, and (c) a request to reconsider their classification. Both agents then produce revised outputs.
5) **Final Arbitration:** The orchestrator evaluates the revised outputs from both agents, weighing them according to their trust profiles and the coherence

of their updated justifications. It then produces a final classification decision along with a meta-level explanation synthesizing the reasoning from both agents and the retrieval evidence.

This hierarchical design ensures that the orchestrator acts as a supervisory layer that synthesizes information from multiple independent agents rather than being an additional agent at the same level. The orchestrator's role is purely meta-cognitive: it does not perceive visual data but evaluates and integrates agent-level outputs to arrive at a trustworthy final decision.

2) TRUST EVALUATION LAYER

The Trust Evaluation Layer conducts offline trust profiling of the AI agents by quantifying their calibration, discriminative reliability, and consistency over a labeled dataset. Specifically, both agents were evaluated in a zero-shot inference setting on the training image set originally employed in Experiment II for fine-tuning, but here presented without ground-truth labels during evaluation. For each image, we recorded the predicted class, associated confidence score, and explanatory output provided by the agent. Using this comprehensive log of model behavior, we computed a suite of quantitative trustworthiness metrics to assess each agent's predictive confidence alignment and overall reliability under zero-shot conditions.

- **Expected Calibration Error (ECE):** In this study, ECE quantifies the discrepancy between a model's predicted confidence and its actual accuracy across prediction bins, serving as a key metric for assessing trustworthiness in zero-shot and fine-tuned Agentic AI systems. A low ECE value indicates that the model's self-reported confidence aligns closely with its empirical correctness, enabling more reliable orchestration decisions based on agent trustworthiness. By incorporating ECE into the orchestrator's arbitration logic, we enhance the system's ability to identify overconfident failures and trigger re-evaluation loops when agent confidence is not well-calibrated.

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right|,$$

  Where B_m is the m-th confidence bin, n the number of samples, and acc and conf represent per-bin accuracy and confidence.
- **Overconfidence Ratio (OCR):** The OCR measures the proportion of incorrect predictions made with high confidence (e.g., confidence > 0.9), capturing the extent to which models exhibit unjustified certainty. A high OCR value signals a critical misalignment between model confidence and actual correctness, often leading to erroneous decisions in confidence-based orchestration pipelines. In our trust-aware framework, OCR serves as a penalizing factor in agent arbitration, allowing the orchestrator to detect and downweight predictions from agents that frequently make wrong decisions with high confidence.

$$\mathrm{OCR} = \frac{|\{i : \hat{y}_i \neq y_i \wedge c_i > 0.9\}|}{|\{i : c_i > 0.9\}|},$$

  This metric measures how often models are confidently wrong.
- **Consistency Gap (CG):** The CG quantifies the divergence in a model's predictions when exposed to semantically equivalent but syntactically varied prompts, serving as a proxy for reasoning stability under linguistic perturbations. A large CG indicates that the AI agent's output is highly sensitive to prompt phrasing, undermining reliability and reproducibility in decision-making. Within our framework, CG is used as a trust signal to evaluate the robustness of agent responses, enabling the orchestrator to identify models prone to inconsistency and initiate re-evaluation when necessary.

$$\mathrm{CG} = \frac{1}{n} \sum_{i=1}^{n} \mathbb{I}[P_i^{(1)} \neq P_i^{(2)}],$$

  This metric assesses variation in outputs across prompt formulations.

This quantitative profiling enabled agent-level trust estimation: a method for assessing an agent's general reliability independently of individual inputs. The orchestrator uses these trust metrics (ECE, OCR, and CG) to understand how much it can trust each agent, thereby modulating their influence in decision fusion and granting higher weight to agents with stronger calibration and lower overconfidence.

3) IMAGE RAG

To complement the trust-aware orchestration mechanism, we incorporated a multimodal retrieval-augmented generation (Image-RAG) system designed for plant disease detection via semantic similarity and weighted voting. This framework utilizes a pre-trained vision-language model (CLIP, ViT-B/32 variant) to embed input images into a high-dimensional feature space that supports interpretable and context-rich decision-making.

Given an input image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$, the CLIP vision encoder transforms it into a 512-dimensional embedding $\mathbf{e} \in \mathbb{R}^{512}$ using a vision transformer architecture. This process includes patch embedding (32 × 32), positional encoding, multi-head self-attention, and a class token aggregation mechanism. Embeddings are subsequently L_2-normalized to lie on the unit hypersphere, enabling cosine similarity to serve as the primary similarity metric:

$$\mathbf{e}_i = \mathrm{L2_normalize}(\mathrm{ViT}_{\mathrm{CLIP}}(\mathbf{I}_i)), \tag{1}$$

$$\mathrm{sim}(\mathbf{e}_i, \mathbf{e}_j) = \cos(\theta_{ij}) = \mathbf{e}_i \cdot \mathbf{e}_j \tag{2}$$

All reference embeddings are stored in a vector database implemented using Facebook AI Similarity Search (FAISS), employing an exact inner product index (`IndexFlatIP`).

Each entry is paired with a category label $y_i \in$ {healthy, black-rot, rust, scab} and associated metadata (e.g., image URLs, index references).

At inference time, the system performs k-nearest neighbor retrieval. Given a query image $\mathbf{I}_q$, we compute its embedding $\mathbf{e}_q$ and retrieve the top-k most similar images:

$$R_k = \arg\max_k \{\cos(\mathbf{e}_q, \mathbf{e}_i) \mid i \in [1, N]\} \tag{3}$$

Each retrieved item is assigned a similarity score $s_i = \cos(\mathbf{e}_q, \mathbf{e}_i)$.

To classify the query image, the system applies a weighted voting mechanism in which the confidence for each category c is computed by normalizing the similarity-weighted votes from the retrieved examples:

$$\text{conf}(c) = \frac{\sum_{i \in R_k, y_i = c} s_i}{\sum_{j \in R_k} s_j} \tag{4}$$

This mechanism produces interpretable confidence scores that reflect both the quantity and quality of visual evidence for each class.

Implementation details include batch processing for efficiency, robust error handling (e.g., corrupted images, missing URLs), and database persistence using FAISS binary formats and Python serialization. The flat index provides exact search with $O(N)$ query complexity, sufficient for moderate dataset sizes. For larger-scale deployment, approximate nearest neighbor indexing may be integrated.

When queried, the Image-RAG system returns a ranked list of candidate categories along with normalized confidence scores, e.g.:

```
[
  {"category": "scab", "confidence": 0.5005},
  {"category": "healthy", "confidence": 0.3996},
  {"category": "rust", "confidence": 0.0999}
]
```

This interpretability and modularity make Image-RAG a natural supplement to the broader Agentic AI classification pipeline.

All source code, algorithm and data are publicly available at https://github.com/Applied-AI-Research-Lab/Orchestrator-Agent-Trust

III. RESULTS

A. ZERO-SHOT AGENTIC CLASSIFICATION WITH CONFIDENCE-AWARE ORCHESTRATION

In the first experimental configuration, we evaluated the baseline performance of a modular Agentic AI system in a zero-shot classification setting. Two general-purpose multimodal agents, GPT-4o and Qwen-2.5-VL, were deployed without any task-specific fine-tuning. Each agent received an image prompt and independently produced a classification label, a natural language rationale, and a normalized confidence score. These outputs were then passed to a non-visual reasoning orchestrator (o3-mini-2025-01-31), which performed structured comparative reasoning to synthesize a final classification decision. The orchestrator operated under a confidence-aware strategy, weighting agent responses based on reported confidence and justification coherence, without any external retrieval or trust calibration mechanisms. This configuration establishes a foundational benchmark to assess model reliability, ensemble effectiveness, and overconfidence dynamics in open-domain zero-shot inference.

The performance metrics of the agents and orchestrator are summarized in Table 1. GPT-4o achieved the highest zero-shot accuracy at 56.88%, followed by the orchestrator at 48.13% and Qwen at 45.00%. Confidence score distributions (Fig. 3a) revealed significant disparities in self-reported certainty, with Qwen averaging 94.3% confidence versus GPT-4o's 87.4% (Table 2). However, this confidence was not reliably predictive of correctness; overconfident misclassifications were common, particularly for Qwen, contributing to a higher OCR. The orchestrator's comparative logic led to a marginal calibration improvement, though the resulting accuracy remained below 50%.

Figure 3b reports top-1 classification accuracy for each agent, where GPT-4o clearly outperforms the others. Weighted precision scores are illustrated in Fig.3c, and raw confidence distributions are presented in Fig.3d. Average confidence levels per model are plotted in Fig.3e, with Qwen showing systematic overconfidence. Recall and F1 scores are detailed in Figs.3f and 3g, respectively, showing GPT-4o leading across metrics.

Confusion matrices in Figs. 3h, 3i, and 3j display inter-class prediction patterns for Qwen, GPT, and the orchestrator, respectively. Misclassification frequently occurred between visually similar disease classes (e.g., black-rot and rust), highlighting the inherent difficulty of this task for zero-shot models.

These findings reveal that while zero-shot multimodal agents can produce fluent and confident predictions, their self-reported confidence often fails to align with empirical performance. Qwen's overconfidence produced high OCR values and unstable outcomes, while GPT-4o demonstrated slightly better alignment between confidence and correctness. The orchestrator, despite lacking visual input, added interpretability and decision consistency through structured reasoning. However, the marginal performance gains and misalignment of confidence and accuracy underscore the need for deeper trust calibration. These limitations served as the motivation for the next experimental stages introducing supervised fine-tuning (Experiment II) and a full trust-aware orchestration with re-evaluation mechanisms (Experiment III).

B. FINE-TUNED AGENTIC MODELS WITH CONFIDENCE-AWARE ORCHESTRATION

To evaluate whether supervised adaptation improves model reliability and ensemble synergy, we conducted a second experiment in which both GPT-4o and Qwen-2.5-VL were fine-tuned on the apple disease classification dataset. The overall setup mirrored Experiment I, retaining identical multimodal inputs, test prompts, and classification

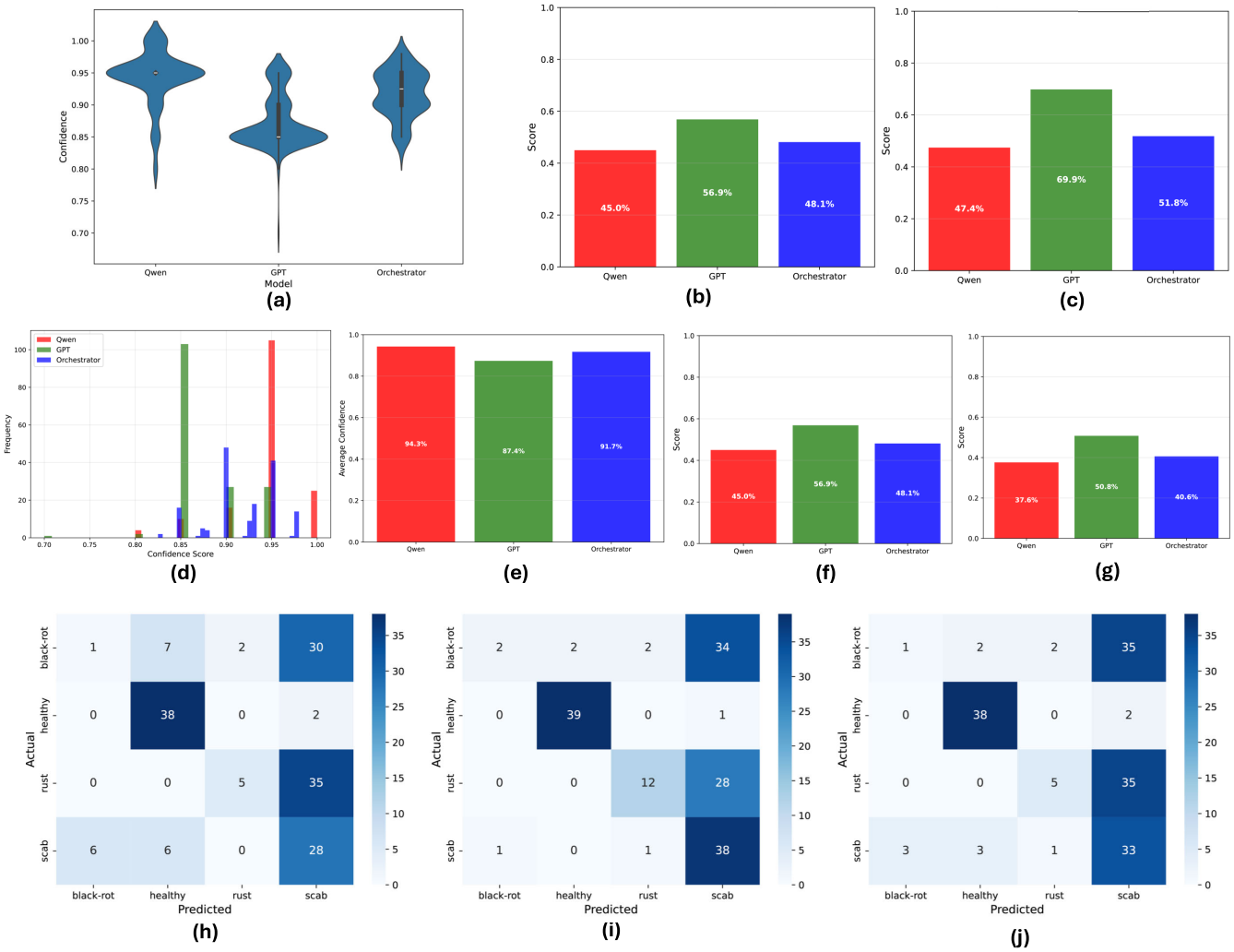

FIGURE 3. Experiment I – Zero-shot performance.(a) Violin plot of model confidence distributions across all classes for Qwen-2.5-VL, GPT-4o, and the orchestrator in the zero-shot configuration (Experiment I). (b) Top-1 classification accuracy comparison of Qwen, GPT, and the orchestrator in Experiment I. (c) Weighted precision scores of all agents in the zero-shot setting. (d) Histogram of raw confidence scores reported by Qwen, GPT, and the orchestrator. (e) Mean confidence scores per agent, aggregated across the entire test set. (f) Weighted recall for each agent in zero-shot inference. (g) Weighted F1-scores for Qwen, GPT, and the orchestrator, highlighting performance balance. (h) Confusion matrix for Qwen-2.5-VL predictions on the evaluation set. (i) Confusion matrix for GPT-4o predictions, revealing inter-class confusion patterns. (j) Confusion matrix for the final orchestrator decision outcomes based on trust-aware arbitration.

objectives. However, the zero-shot agents were replaced with domain-adapted variants, each fine-tuned on a labeled training set of 512 samples and validated on 128 additional samples. Importantly, the orchestrator (o3-mini-2025-01-31) remained unchanged and continued to perform confidence-aware arbitration, relying solely on agent-generated confidence scores and textual justifications to make final decisions.

This design tests whether fine-tuning alone, without any updates to the orchestration logic or external retrieval modules, could significantly enhance prediction accuracy, calibration, and inter-agent agreement. Table 1 presents the updated metrics: GPT-4o achieved an accuracy of 98.13%, while Qwen-2.5-VL reached 95.63%. The orchestrator, despite lacking direct visual input, attained 97.50% accuracy, underscoring the impact of improved agent outputs on system-wide arbitration. Weighted and macro-averaged precision, recall, and F1 scores exceeded 95% for all models, indicating a substantial leap from the zero-shot performance baseline.

Figure 4a shows violin plots of agent confidence distributions after fine-tuning. Compared to the zero-shot setting, these distributions are narrower and more centered, suggesting improved calibration. Figure 4b reports top-1 accuracy, while Fig.4c highlights gains in weighted precision across agents. Notably, the confidence histograms in Fig.4d exhibit reduced overconfident noise, and the average confidence values in Fig. 4e show a modest increase (e.g., GPT-4o rose from 87.4% to 92.58%), indicating better alignment between certainty and correctness (Table 4).

TABLE 1. Performance metrics (accuracy, weighted and macro precision, recall, and F1) for each agent across three experimental configurations. All metrics for each experiment are shown in the same row.

Experiment	Qwen	GPT	Orchestrator
I. Zero-Shot + Confidence-Aware	Accuracy: 0.4500	Accuracy: 0.5688	Accuracy: 0.4813
	Precision$_w$: 0.4742	Precision$_w$: 0.6985	Precision$_w$: 0.5183
	Recall$_w$: 0.4500	Recall$_w$: 0.5688	Recall$_w$: 0.4813
	F1-score$_w$: 0.3763	F1-score$_w$: 0.5078	F1-score$_w$: 0.4062
	Precision$_m$: 0.4742	Precision$_m$: 0.6985	Precision$_m$: 0.5183
	Recall$_m$: 0.4500	Recall$_m$: 0.5688	Recall$_m$: 0.4813
	F1-score$_m$: 0.3763	F1-score$_m$: 0.5078	F1-score$_m$: 0.4062
II. Few-Shot + Confidence-Aware	Accuracy: 0.9563	Accuracy: 0.9813	Accuracy: 0.9750
	Precision$_w$: 0.9603	Precision$_w$: 0.9817	Precision$_w$: 0.9765
	Recall$_w$: 0.9563	Recall$_w$: 0.9813	Recall$_w$: 0.9750
	F1-score$_w$: 0.9558	F1-score$_w$: 0.9812	F1-score$_w$: 0.9747
	Precision$_m$: 0.9603	Precision$_m$: 0.9817	Precision$_m$: 0.9765
	Recall$_m$: 0.9563	Recall$_m$: 0.9813	Recall$_m$: 0.9750
	F1-score$_m$: 0.9558	F1-score$_m$: 0.9812	F1-score$_m$: 0.9747
III. Zero-Shot + Trust-Aware + RAG	Accuracy: 0.7313	Accuracy: 0.8750	Accuracy: 0.8563
	Precision$_w$: 0.7526	Precision$_w$: 0.8898	Precision$_w$: 0.8591
	Recall$_w$: 0.7313	Recall$_w$: 0.8750	Recall$_w$: 0.8563
	F1-score$_w$: 0.7292	F1-score$_w$: 0.8690	F1-score$_w$: 0.8555
	Precision$_m$: 0.7526	Precision$_m$: 0.8898	Precision$_m$: 0.8591
	Recall$_m$: 0.7313	Recall$_m$: 0.8750	Recall$_m$: 0.8563
	F1-score$_m$: 0.7292	F1-score$_m$: 0.8690	F1-score$_m$: 0.8555

Weighted (subscript w) and macro-averaged (subscript m) precision, recall, and F1 scores are reported.

Recall and F1 improvements are presented in Figs. 4f and 4g, respectively. The F1 score for GPT-4o increased from 50.78% to 98.12%, and for Qwen from 37.63% to 95.58%. These gains reveal that fine-tuning not only enhances correct predictions but also reduces false positives and negatives. Confusion matrices in Figs. 4h–4j show sharply improved class-level discrimination. Notably, categories like black-rot and rust, which previously showed misclassification overlap, now exhibit strong diagonal dominance.

The orchestrator benefited substantially from the improved agent outputs. Since its decision logic depends on comparative evaluation of confidence-aligned explanations, the increase in calibration quality directly translates to more reliable ensemble predictions. Interestingly, although the orchestrator does not receive image inputs, its final predictions align well with the dominant and better-calibrated agent in most cases.

The results confirm that supervised adaptation significantly enhances the classification performance of general-purpose vision-language agents within the agentic AI framework. After fine-tuning on the domain-specific apple disease dataset, both GPT-4o and Qwen-2.5-VL showed notable gains across accuracy, precision, recall, and F1 scores. As shown in Table 1, GPT-4o achieved 98.13% accuracy and Qwen 95.63%, while the orchestrator despite lacking direct image access reached 97.50% accuracy. This highlights how confidence-aware arbitration benefits from well-calibrated agent responses, with the orchestrator's performance reflecting the value of alignment between self-reported confidence and correctness.

TABLE 2. Experiment I: Confidence score distribution.

Statistic	Qwen	GPT-4o	Orchestrator
Count	160	160	160
Mean	0.943	0.874	0.917
Std. Dev.	0.042	0.042	0.037
Min	0.800	0.700	0.825
25th Pctl	0.950	0.850	0.900
Median	0.950	0.850	0.925
75th Pctl	0.950	0.900	0.950
Max	1.000	0.950	0.980

However, the experiment also revealed a key limitation: when both fine-tuned agents confidently agree on an incorrect prediction, the orchestrator lacks a dissenting signal and cannot intervene. This consensus failure mode exposes the risk of overreliance on internal confidence and justifications alone. To address such blind spots, the next experimental framework integrates external trust metrics and retrieval-based verification. These enhancements aim to assess not only what agents predict, but also how trustworthy and evidence-grounded those predictions are especially under ambiguity or in high-stakes settings.

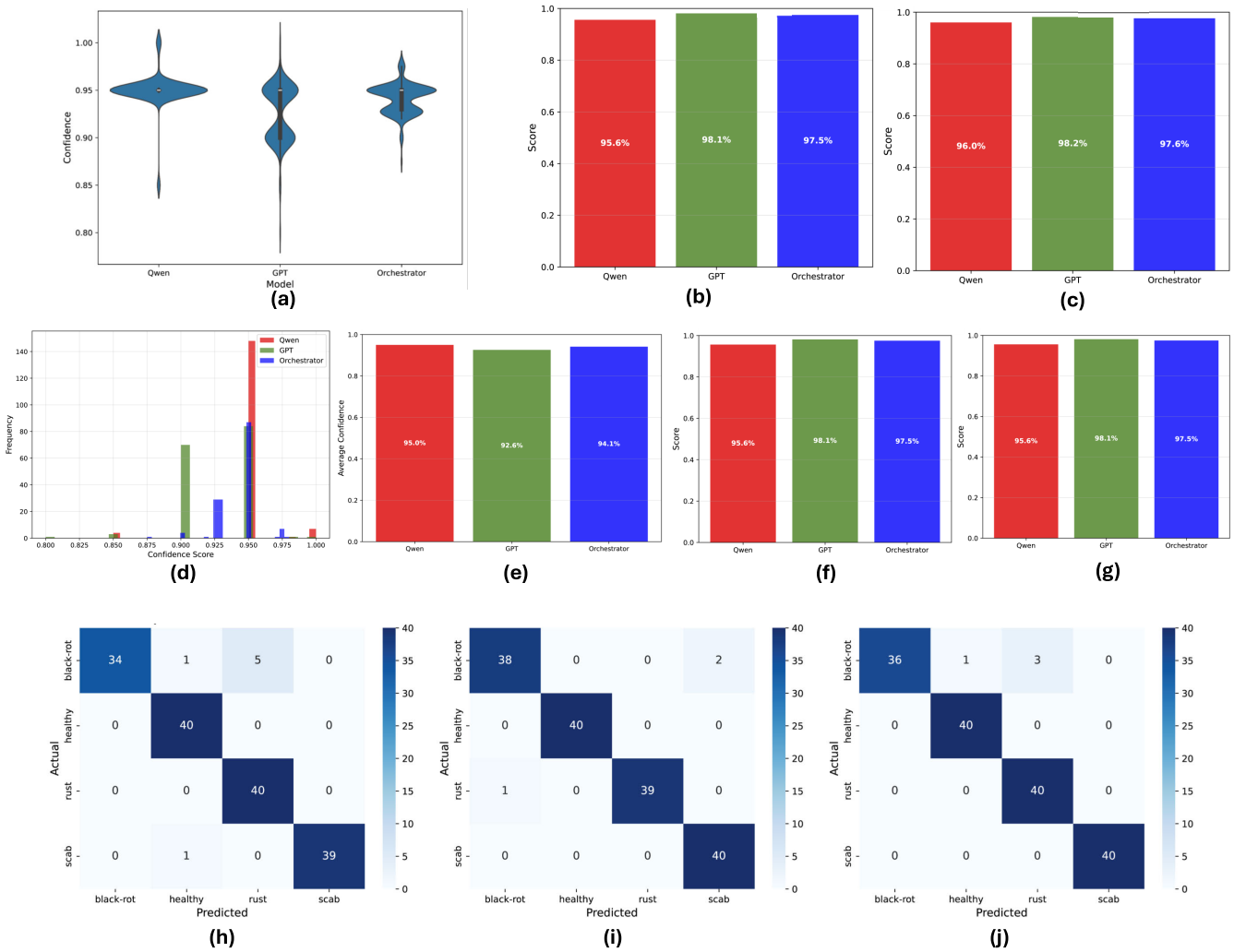

FIGURE 4. Experiment II – Fine-tuned agents.(a) Violin plot of confidence distributions across all classes for Qwen-2.5-VL, GPT-4o, and the orchestrator following supervised fine-tuning.(b) Top-1 classification accuracy comparison after agent fine-tuning.(c) Weighted precision scores across the three agents, showing enhanced discriminative performance.(d) Histogram of raw confidence scores across all predictions, reflecting sharper calibration post fine-tuning.(e) Mean confidence scores per agent, showing convergence in self-reported certainty.(f) Weighted recall scores for Qwen, GPT, and the orchestrator on the test set.(g) Weighted F1-scores highlighting overall performance balance improvements.(h) Confusion matrix for Qwen-2.5-VL showing improved inter-class discrimination.(i) Confusion matrix for GPT-4o illustrating reduced misclassification frequency.(j) Confusion matrix for the orchestrator's final decisions, demonstrating stability in arbitration after fine-tuning.

TABLE 3. Experiment II: GPT-4o Fine-tuning comparison.

Config	Epochs	Batch	Val Loss	Time (s)	Cost ($)
ResNet-50 tuned	10	16	0.0088	1778	47.53
Default settings	3	1	0.0617	1652	13.09

TABLE 4. Experiment II: Confidence statistics.

Statistic	Qwen	GPT-4o	Orchestrator
Count	160	160	160
Mean	0.950	0.926	0.941
Std. Dev.	0.019	0.029	0.016
Min	0.85	0.80	0.875
25th Pctl	0.95	0.90	0.93
Median	0.95	0.95	0.95
75th Pctl	0.95	0.95	0.95
Max	1.00	1.00	0.98

C. TRUST-AWARE ORCHESTRATION WITH RAG AND RE-EVALUATION LOOPS

To overcome the reliability ceiling observed in previous settings, the third experiment introduces a trust-aware orchestration framework augmented with a RAG pipeline. Unlike Experiment II, this configuration retains the zero-shot GPT-4o and Qwen-2.5-VL agents to investigate whether reasoning and calibration improvements can be realized without task-specific fine-tuning. Enhancements to the orchestrator include: (i) a multi-metric trust scoring module incorporating ECE, OCR, and CG, and (ii) a dynamic Re-Evaluation Loop, triggered when agent trust scores fall below a learned threshold.

In this setup, low-trust predictions trigger the orchestrator to initiate a re-evaluation loop, prompting agents with their

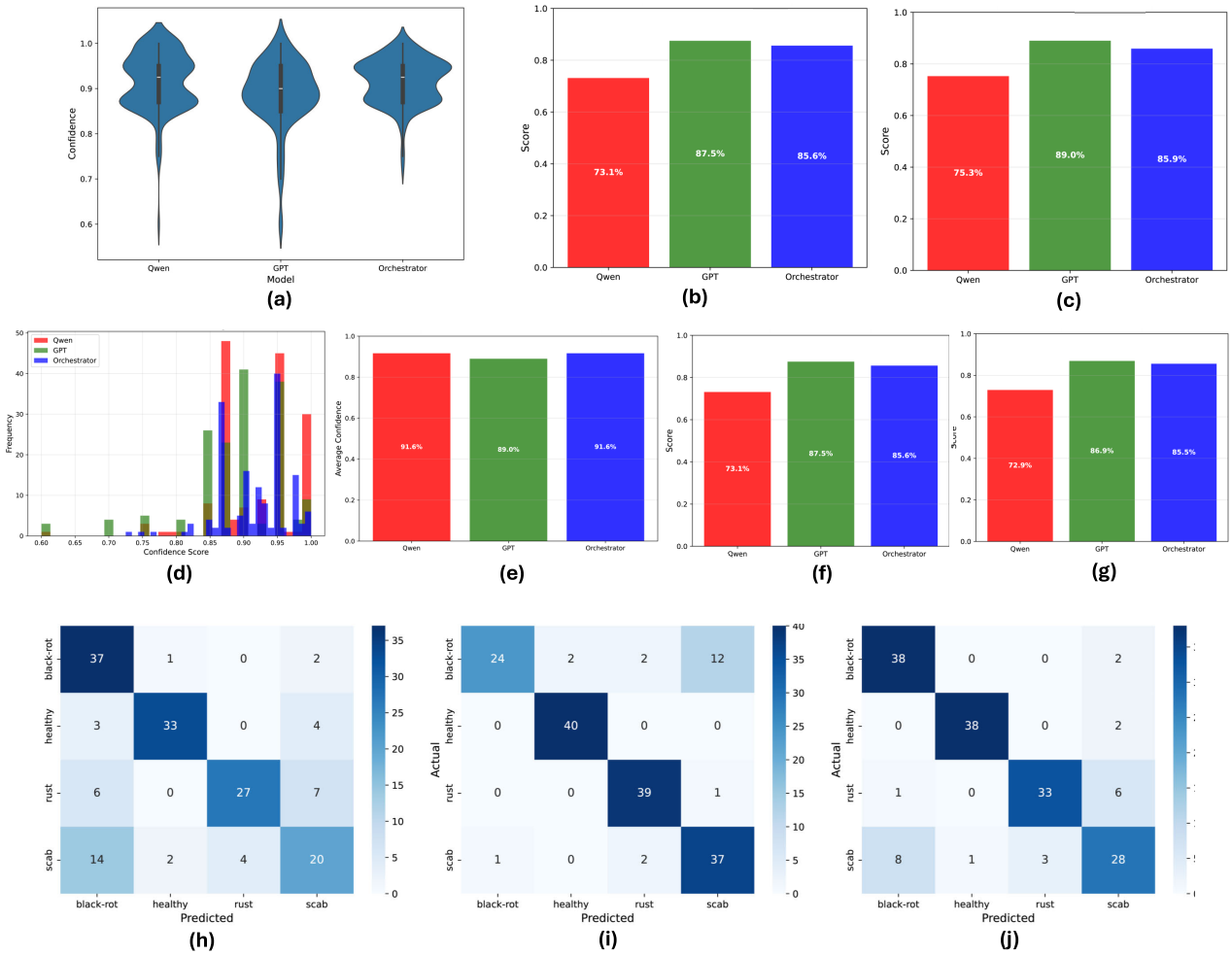

FIGURE 5. Experiment III – Trust-Aware Orchestration with RAG and Re-Evaluation Loops.(a) Violin plot depicting confidence distributions across all classes for Qwen-2.5-VL, GPT-4o, and the orchestrator in the trust-aware setup with retrieval augmentation and re-evaluation. (b) Accuracy comparison across the three agents under trust-calibrated arbitration. (c) Weighted precision scores for all models following trust-aware reasoning. (d) Histogram of raw confidence outputs from each agent after trust score filtering. (e) Mean confidence values per agent post re-evaluation, showing enhanced calibration and reduced overconfidence. (f) Weighted recall metrics across all models in the final pipeline. (g) Weighted F1-scores reflecting the balance of precision and recall under trust-informed decision-making. (h) Confusion matrix for Qwen-2.5-VL predictions after Image-RAG integration and re-evaluation. (i) Confusion matrix for GPT-4o responses within the trust-aware system. (j) Final decision confusion matrix of the orchestrator, highlighting improvements in accuracy and reduced inter-class confusion due to trust filtering and context-grounded retrieval.

TABLE 5. Experiment III: Trust metrics for zero-shot agents.

Model	Acc.	Avg Conf	$\text{Conf}_{\text{corr}}$	$\text{Conf}_{\text{incorr}}$	CG	OCR	HCW	THC	CCC	p-val	ECE	CWA
Qwen	0.492	0.945	0.950	0.941	0.009	0.508	260	512	0.126	0.0042	0.453	0.495
GPT	0.584	0.877	0.890	0.860	0.030	0.416	213	512	0.361	0.0000	0.293	0.592

Note: Acc.=Accuracy; OCR=Overconfidence Ratio; HCW=High-confidence wrong; THC=Total high-confidence; CCC=Confidence-Correctness Correlation; ECE=Expected Calibration Error; CWA=Confidence-Weighted Accuracy.

Note: Acc.=Accuracy; OCR=Overconfidence Ratio; HCW=High-confidence wrong; THC=Total high-confidence; CCC=Confidence-Correctness Correlation; ECE=Expected Calibration Error; CWA=Confidence-Weighted Accuracy.

prior responses and augmenting the input with semantic retrievals from Image-RAG. The retrieval module uses CLIP (ViT-B/32) to encode a curated set of class-representative reference images into 512-dimensional embeddings, which are L_2-normalized and stored in a FAISS vector database using an exact inner product index (`IndexFlatIP`). These reference images are sourced from the same training set (512 labeled images across the four disease categories) used in Experiment II for fine-tuning, ensuring that the retrieval database contains domain-specific visual exemplars with verified ground-truth labels. Each reference embedding is associated with a disease category label $y_i \in \{\text{healthy, black-rot, rust, scab}\}$ and enriched with metadata such as image URLs and textual definitions. During re-evaluation, the agents receive the top-k

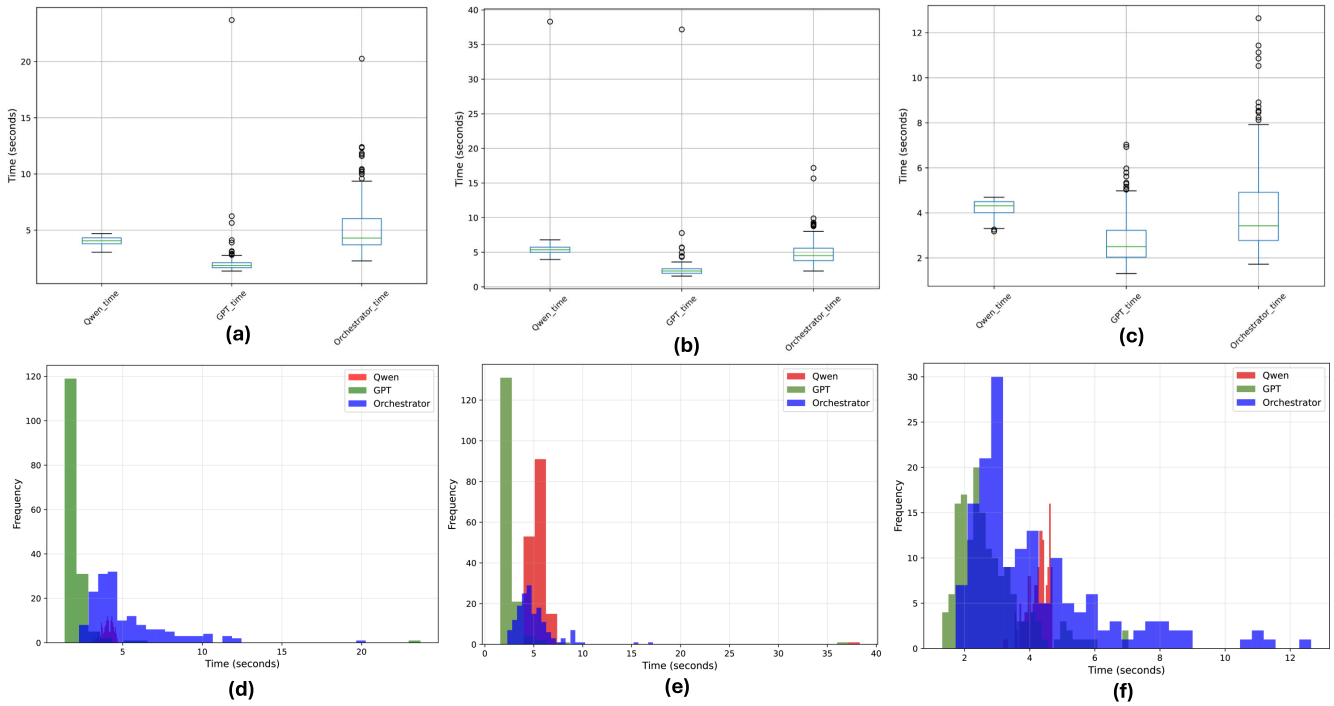

FIGURE 6. Time performance analysis across experimental configurations.a) Boxplot showing inference time distribution per image in Experiment I (zero-shot setting). b) Boxplot showing inference time distribution in Experiment II (fine-tuned setting). c) Boxplot showing inference time distribution in Experiment III (trust-aware orchestration with RAG). d) Histogram of inference time frequencies for Experiment I. e) Histogram of inference time frequencies for Experiment II. f) Histogram of inference time frequencies for Experiment III. These visualizations highlight how orchestration strategies and model configurations affect latency, offering insight into the computational trade-offs of Agentic AI systems.

TABLE 6. Experiment III: Confidence after re-evaluation.

Statistic	Qwen	GPT-4o	Orchestrator
Count	160	160	160
Mean	0.916	0.890	0.916
Std. Dev.	0.064	0.074	0.050
Min	0.60	0.60	0.725
25th Pctl	0.87	0.85	0.87
Median	0.925	0.90	0.925
75th Pctl	0.95	0.95	0.95
Max	1.00	1.00	1.00

TABLE 7. Top 5 Hyperparameter configurations for Qwen-2.5-VL.

Rank	Trial	Val Loss	LR	Batch	Warmup	Epochs
1	11	0.000010	1.094×10^{-5}	4	0.0997	15
2	16	0.000934	1.901×10^{-5}	4	0.0881	13
3	15	0.000993	2.348×10^{-5}	4	0.0864	12
4	9	0.001013	1.552×10^{-4}	2	0.0351	8
5	19	0.001188	1.720×10^{-5}	4	0.0898	12

most similar reference examples based on cosine similarity, integrated into an updated prompt. This guides them to revise or reaffirm their prediction with additional visual and semantic context. The orchestrator then assesses the updated responses using trust metrics (ECE, OCR, CG) and finalizes the decision based on the most coherent and trustworthy agent output.

1) PERFORMANCE

As summarized in Table 1, GPT-4o achieved 87.50% accuracy and Qwen reached 73.13%, both significantly outperforming their initial zero-shot baselines in Experiment I. The orchestrator attained an accuracy of 85.63%, demonstrating that trust-aware orchestration, combined with retrieval-based contextualization, can close a large portion of the performance gap typically addressed via fine-tuning. Subfigure 5b shows the accuracy improvements, while subfigures 5c, 5f, and 5g illustrate the precision, recall, and F1-score gains, respectively.

2) CALIBRATION

As shown in Fig. 5a, the violin plot of confidence distributions revealed a reduction in overconfidence for both agents. Mean confidence scores (Fig. 5e) slightly decreased compared to Experiment II, suggesting more cautious, better-calibrated outputs. The orchestrator's histogram (Fig. 5d) reflects this calibrated behavior, with lower variance and reduced extremity in self-reported confidence (Table 6).

3) QUALITATIVE GAINS

Confusion matrices shown in Figs. 5h–j provide insight into class-specific performance. RAG and trust-based re-evaluation mitigated frequent misclassifications seen in Experiment I especially in visually ambiguous classes such as black-rot and rust by incorporating external

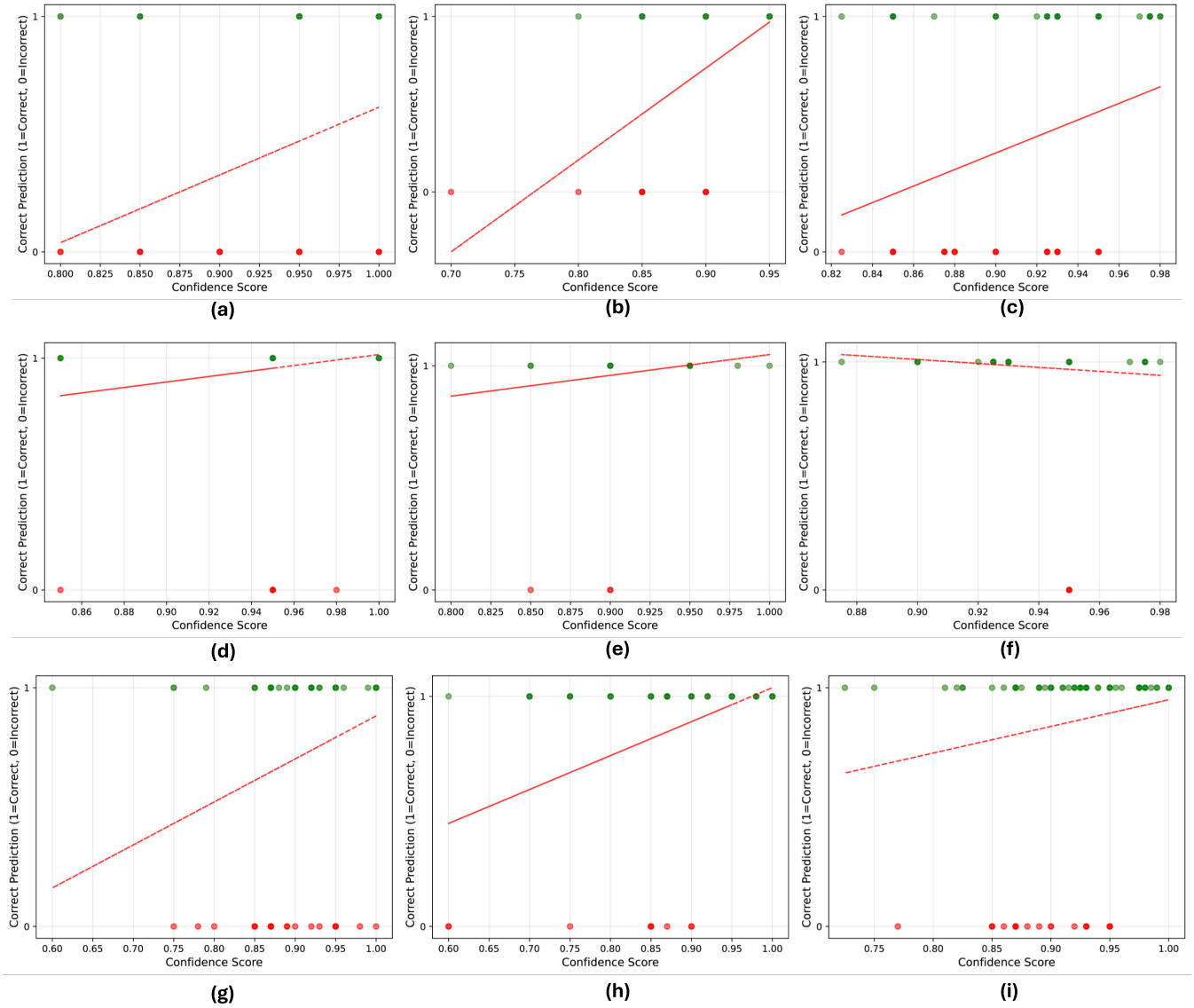

FIGURE 7. Confidence vs. accuracy calibration analysis across experiments.a) Confidence vs. accuracy plot for Qwen-2.5-VL in Experiment I (zero-shot setting);b) Confidence vs. accuracy plot for GPT-4o in Experiment I;c) Confidence vs. accuracy plot for the orchestrator in Experiment I;d) Confidence vs. accuracy plot for Qwen-2.5-VL in Experiment II (fine-tuned setting);e) Confidence vs. accuracy plot for GPT-4o in Experiment II;f) Confidence vs. accuracy plot for the orchestrator in Experiment II;g) Confidence vs. accuracy plot for Qwen-2.5-VL in Experiment III (trust-aware orchestration with RAG);h) Confidence vs. accuracy plot for GPT-4o in Experiment III;i) Confidence vs. accuracy plot for the orchestrator in Experiment III. These calibration curves illustrate the alignment between predicted confidence and true accuracy across agents and experimental conditions, highlighting changes in overconfidence and calibration quality. The solid red line indicates the line of best fit (observed trend) for the model's predictions, while the dashed diagonal line (where visible) represents perfect calibration.

visual-textual anchors. Notably, GPT-4o showed strong diagonal dominance in its matrix, and the orchestrator successfully avoided the propagation of low-trust predictions even when both agents initially made incorrect predictions.

4) TRUST PROFILING

From the profiling, several insights emerge regarding agent reliability under zero-shot conditions. Importantly, both models exhibit suboptimal calibration and trustworthiness metrics overall (Table 5) an expected outcome given that they were not fine-tuned on the task-specific dataset. Nonetheless, relative differences offer valuable guidance for orchestration design:

- GPT demonstrates superior calibration, with a lower ECE (ECE = 0.293 vs. 0.453 for Qwen) and a substantially higher Confidence–Correctness Correlation (CCC = 0.361, $p < 0.0001$), suggesting that its confidence estimates are more aligned with empirical correctness and thus more actionable in trust-aware orchestration.
- GPT exhibits greater reliability under uncertainty, evidenced by a lower Overconfidence Rate (OCR = 0.416 vs. 0.508) and fewer high-confidence errors

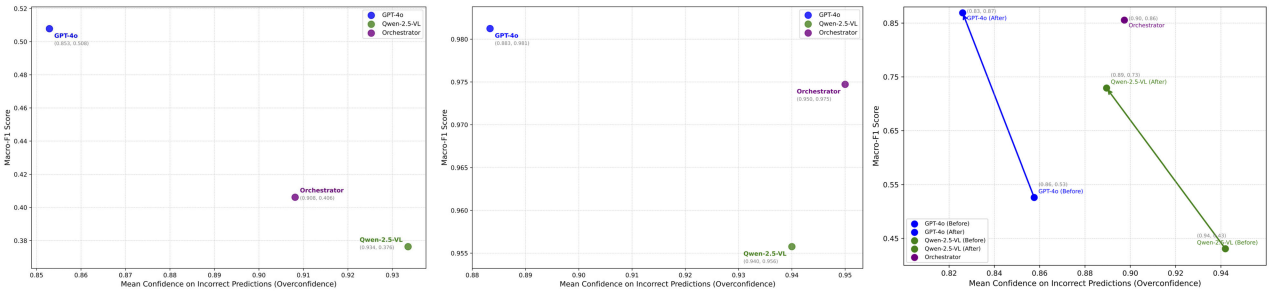

FIGURE 8. Agent overconfidence vs. final macro-F1.(a) Experiment I: zero-shot predictions – overconfidence vs. macro-F1 score;(b) Experiment II: few-shot predictions – overconfidence vs. macro-F1 score;(c) Experiment III: trust-aware orchestration with RAG – overconfidence vs. macro-F1 score, both before and after re-evaluation loop.

```
"agent_prompt": """Analyze the provided image of an apple leaf using your
  computer vision capabilities. Classify the leaf into the most appropriate
  category based on its condition, choosing from the predefined list:
  {
  "categories": [
    "black-rot",
    "healthy",
    "rust",
    "scab"
    ]
  }
  Provide your final classification in the following JSON format without
  explanations:
  {
    "category": "chosen_category_name",
    "reason": "brief_reason_for_this_classification",
    "confidence": "confidence_score_between_0_and_1"
  }""",

"agent_response": {'category': 'scab',
                   'reason': 'The presence of brown spots and scarring on the'
                   'leaf surfaceis characteristic of scab disease.',
                   'confidence': 0.95, 'time': 4.0999331474, 'cost': 0.001035}
```

FIGURE 9. AI agent prompt and response for disease classification. The prompt instructs a vision-language model to classify apple leaf diseases. The response includes the predicted category, justification, self-reported confidence, latency, and estimated computational cost in JSON format.

(213 vs. 260), indicating a reduced risk of confidently incorrect predictions a critical feature for high-stakes decision environments.

- Qwen displays overconfidence and poor discrimination, reporting the highest average confidence (0.945) but a minimal Confidence Gap (CG = 0.009), meaning it struggles to differentially calibrate its confidence for correct versus incorrect predictions. This weakens its applicability in systems that depend on confidence signals for agent weighting or override logic.

These findings enabled the orchestrator to weight GPT's predictions more heavily during ensemble decision-making, thereby improving the robustness and accuracy of the overall Agentic AI system.

5) DISAGREEMENT ANALYSIS AND TRUST ARBITRATION

An analysis of disagreements between agents and system components provides further insight: In 12.5% of cases, the GPT agent refused the orchestrator's recommendation for re-evaluation, returning the same prediction as before. However, in only 3 out of those 20 cases did this result in a correct prediction. The same pattern, but to a greater extent, was observed with Qwen: the agent ignored the re-evaluation recommendation in 29.38% of cases, returning the same prediction. Yet only 5 out of 47 such instances led to a correct outcome.

```
"orchestrator_prompt": """You are acting as the orchestrator in an agentic AI
  system. Your role is to make an informed and well-reasoned final decision
  for a classification task, based on the outputs of multiple AI agents.
  Task:
  Classify an image of an apple plant leaf into the correct disease category.
  Context:
  You have received independent responses from two specialized AI agents.
  Each agent may use different modalities, models, or reasoning approaches.
  Your responsibility is to analyze, compare, and synthesize their outputs to
  make a final, justified classification decision.

  $agents_response

  Your Task as Orchestrator:
  Based on the agents' responses, provide your final classification decision.

  You must return your answer in the following JSON format:
  {
    "category": "chosen_category_name",
    "reason": "detailed explanation of your reasoning process",
    "confidence": "confidence_score_between_0_and_1"
  }

  Be sure your reasoning includes:
  - How you evaluated the agents' outputs
  - Any conflicts or agreements you observed
  - The rationale behind your chosen decision"""

"orchestrator_response": {'category': 'scab', 'reason': 'Both AI Agent 1 and'
  'AI Agent 2 identified the disease as scab based on the presence of dark,'
  'scabby spots and characteristic brown spots with scarring on the leaf'
  'surface. Given the strong agreement between both agents, and their high'
  'confidence levels (0.95 and 0.85), the evidence clearly points to scab'
  'disease. This consensus, along with the corroborating descriptions, supports'
  'a reliable classification.',
  'confidence': 0.93, 'time': 3.1174688339, 'cost': 0.000787}
```

FIGURE 10. Orchestrator prompt and agentic response for decision arbitration. The orchestrator receives outputs from multiple agents and synthesizes them into a single trusted decision. The JSON response includes the class, rationale, confidence, processing time, and cost.

These low correctness rates when agents disregard re-evaluation prompts underscore the value of the hierarchical architecture: while the re-evaluation loop and Image-RAG provide critical external evidence, optimal outcomes are achieved when the orchestrator functioning as the meta-reasoning layer performs the final arbitration, synthesizing information from both the agents' revised outputs and the retrieval context to make an informed trust-calibrated decision.

Further analysis of orchestrator-agent disagreements shows:

```
Using RAG...
{'response': [{'category': 'rust', 'confidence': 1.0}], 'time': 0.1972808838,
  'cost': 5e-05}
Using Qwen Unsloth Base Model | Zero-Shot...
{'category': 'scab', 'reason': 'Presence of brown spots that resemble scab'
  'fungal infection.', 'confidence': 0.85, 'time': 3.7545511723,
  'cost': 0.000948}
Using GPT OpenAI Base Model | Zero-Shot...
{'category': 'rust', 'reason': 'Presence of small, rust-colored spots on the'
  'leaf surface.', 'confidence': 0.85, 'time': 2.1049318314, 'cost': 0.001357}
Using GPT Reasoning Model as an Orchestrator...
{'agent_1': 'Please reconsider your selection, taking into account the'
  'Image-RAG. You are not required to change your selection if you are'
  'confident in it. However, if you do change your answer, please explain in'
  'detail why you changed your mind, providing reasoning based on the image.',
'agent_2': 'Please reconsider your selection, taking into account the'
  'Image-RAG. You are not required to change your selection if you are'
  'confident in it. However, if you do change your answer, please explain in'
  'detail why you changed your mind, providing reasoning based on the image.',}
Using Qwen Unsloth Base Model | Zero-Shot...
{'category': 'rust', 'reason': 'The leaf exhibits characteristic spots'
  'indicative of a fungal infection, which aligns closely with the appearance'
  'of rust symptoms.', 'confidence': 0.95, 'time': 4.6233711243,
  'cost': 0.001167}
Using GPT OpenAI Base Model | Zero-Shot...
{'category': 'rust', 'reason': 'The leaf displays small, rust-colored spots'
'consistent with rust disease.', 'confidence': 1.0, 'time': 2.9448902607,
 'cost': 0.001357}
Using GPT Reasoning Model as an Orchestrator...
{'category': 'rust', 'reason': 'Both AI Agent 1 and AI Agent 2 independently'
  'classified the disease as rust based on clear visual indicators such as'
  'characteristic rust-colored spots on the leaf. There is complete agreement'
  'between the agents, with each noting the visible signs of fungal infection'
  'typical of rust disease. The high confidence scores (0.95 and 1.0) further'
  'reinforce the decision to classify the disease as rust.',
  'confidence': 0.975, 'time': 2.5999305248, 'cost': 0.001569}
```

FIGURE 11. Experiment III: Trust-aware orchestration with RAG and re-evaluation. The orchestrator triggers CLIP-based retrieval (Image-RAG) and a re-evaluation loop. Updated agent responses are scored by trust metrics to produce a final decision.

- Orchestrator vs. GPT: Disagreements occurred 36 times (22.5%), with the orchestrator being correct in 16 of those cases (44.44%).
- Orchestrator vs. Qwen: Disagreements occurred 22 times (13.75%), with the orchestrator being correct in 21 cases (95.45%).

The orchestrator significantly outperforms Qwen in disagreement scenarios and even surpasses GPT nearly half the time, underscoring the value of trust-aware arbitration beyond simple majority voting.

However, the re-evaluation process is not without drawbacks. In some instances, it led to overcorrection:

- GPT changed correct answers to incorrect ones 3 times (1.88%) after receiving a re-evaluation prompt.
- Qwen did so 22 times (13.75%).

While rare for GPT, Qwen's frequent revision of correct answers indicates a higher sensitivity to prompt influence, highlighting the need for better confidence calibration and tighter visual-textual integration.

6) INTERPRETABILITY AND SCALABILITY

While the orchestrator achieved near-fine-tuned accuracy without labeled training data, residual failure cases remain, especially when both agents confidently agree on a wrong label. This scenario exposes the limits of current trust metrics to fully capture uncertainty in high-confidence false predictions. Nevertheless, the current framework provides a scalable, modular, and interpretable zero-shot system that balances generalizability and reliability.

Collectively, Experiment III highlights that zero-shot generalist agents, when embedded in trust-calibrated Agentic AI systems with access to external retrieval, can achieve expert-level accuracy in image classification tasks. These findings support a broader vision of scalable agentic intelligence where trust, not tuning, becomes the key to real-world deployment.

D. COMPARATIVE PERFORMANCE AND ABLATION ANALYSIS

1) TIME AND CALIBRATION PERFORMANCE ACROSS CONFIGURATIONS

Figure 6 presents a comparative analysis of inference time across the three experimental setups. Subfigures 6a–c show that fine-tuned agents (Experiment II) achieve the lowest median latency (approximately 1.42 seconds per image) due to optimized internal representations and absence of retrieval overhead, while trust-aware orchestration with retrieval (Experiment III) introduces modest time overhead from CLIP-based retrieval and iterative re-evaluation cycles, resulting in a median inference time of approximately 7.43 seconds per image. Histograms in subfigures 6d-f reveal that Experiment III exhibits a heavier tail in inference time distribution due to variable re-evaluation frequency, with some samples requiring multiple agent consultation rounds. The computational overhead of Experiment III relative to Experiment I (zero-shot baseline) is approximately 1.3×, representing an acceptable trade-off for applications where offline or semi-real-time processing is permissible, such as agricultural diagnostic systems, quality control pipelines, or batch image analysis workflows. While fine-tuned models yield the highest performance with minimal inference cost, they require substantial upfront training investment and lack cross-domain transferability. In contrast, Experiment III achieves near-optimal accuracy (85.6%) without fine-tuning, offering a practical compromise for label-scarce, rapidly evolving, or resource-constrained deployment scenarios where model retraining is infeasible.

2) CONFIDENCE-ACCURACY CALIBRATION AND OVERCONFIDENCE MITIGATION

Figure 7 illustrates confidence–accuracy calibration curves for Qwen-2.5-VL, GPT-4o, and the orchestrator across all experiments. In the zero-shot setting (subfigures 7a–c), both agents display considerable overconfidence, with confidence often exceeding empirical accuracy. This misalignment is partially mitigated by the orchestrator, which arbitrates between agent outputs. Fine-tuning (subfigures 7d–f) improves calibration, especially for GPT-4o, aligning predicted confidence more closely with true correctness. Trust-aware orchestration in Experiment III (subfigures 7g–i) further suppresses overconfidence through calibrated re-evaluation. Figure 8 visualizes the relationship between agent overconfidence (mean confidence on incorrect predictions) and final macro-F1, confirming

that trust-augmented pipelines better align certainty with correctness.

IV. DISCUSSION AND FUTURE DIRECTIONS

A. SUMMARY AND FINDINGS

The increasing complexity and autonomy of AI systems calls for robust, interpretable, and generalizable architectures that can reason, evaluate, and adapt in real time [30]. This study presents a systematic exploration of a novel trust-aware Agentic AI framework that blends zero-shot vision-language agents with orchestration, trust calibration, and RAG-based re-evaluation.

Our three experimental configurations demonstrated that trust-aware orchestration yields up to a 77.94% improvement over confidence-based zero-shot orchestration (from 48.13% to 85.63% accuracy), approaching the performance of fine-tuned agents (97.50%) without requiring task-specific retraining. This is particularly valuable for scaling to large agent populations where per-agent fine-tuning becomes prohibitively expensive. The framework enables seamless integration of new agents in zero-shot settings while the trust-aware orchestrator ensures reliable system-wide performance through calibrated arbitration.

A central insight from this study is the critical role of calibration and trust estimation in Agentic AI systems. While traditional ensemble methods assume independent and reliable agent performance [31], [32], such strategies falter under zero-shot conditions where agents exhibit systematic overconfidence or miscalibration [33], [34]. Our trust-aware orchestration strategy addresses this by incorporating metrics (ECE, OCR, CG) that quantitatively assess confidence-correctness alignment and reasoning consistency. By down-weighting overconfident predictions and triggering re-evaluation when trust is low, the system becomes more resilient to epistemic failures.

From a system design perspective, our architecture prioritizes modularity and plug-and-play scalability, enabling seamless integration of agents with minimal friction. Each agent operates autonomously and can be added, removed, or updated without necessitating retraining of the orchestrator a critical property for real-world deployment where agent capabilities may evolve over time.

B. LIMITATIONS

Several limitations remain. First, although the system outperforms conventional zero-shot baselines, its accuracy still lags behind that of domain-specific, fine-tuned models. This performance gap highlights the trade-off between generalizability and task-specific optimization.

Second, in the zero-shot setting, the design and phrasing of prompts play a critical role in shaping model outputs. Despite clear and structured prompting, we observed that models particularly Qwen-2.5-VL frequently failed to follow the expected response format. To address this, we implemented a re-prompting loop with a capped number of retries, which introduces additional inference overhead and increases system complexity.

Third, the results reported in this study are based on a specific image dataset; therefore, performance on other image-based datasets may vary. However, we expect the relative trends to hold, with fine-tuned models generally outperforming zero-shot approaches and trust-aware orchestration with RAG (Experiment III) providing substantial gains over confidence-based zero-shot orchestration (Experiment I).

Fourth, while we used the o3-mini model as the orchestrator for these experiments, there are several alternative models available. We specifically chose an orchestrator without visual capabilities to avoid biases introduced by its own predictions. Nonetheless, more advanced models such as o3-pro, o4-mini, or other variants with visual understanding could potentially improve orchestration performance.

C. FUTURE PERSPECTIVE

Advancing trust-aware Agentic AI systems presents several promising directions. Incorporating orchestrators with enhanced multimodal reasoning capabilities such as more advanced visual-language models could improve the reliability and fairness of decision arbitration.

Further, optimizing prompt design and exploring adaptive prompting strategies will be essential to address current limitations in zero-shot settings, reducing the need for repeated prompting and improving compliance with response formats.

Scaling the framework to accommodate larger and more diverse populations of AI agents poses challenges related to agent management, specialization, and dynamic trust assessment. Developing methods to efficiently integrate and update agent trust profiles will be critical for maintaining system robustness.

An unexplored but promising configuration would combine fine-tuned agents with trust-aware orchestration and RAG, potentially achieving near-optimal performance while maintaining interpretability and robustness. While Experiment II demonstrated that fine-tuning agents can achieve 97.50% accuracy, we identified a critical failure mode: when both fine-tuned agents confidently agree on an incorrect prediction, the orchestrator lacks a dissenting signal and cannot intervene. Integrating the trust metrics and Image-RAG mechanisms from Experiment III into the fine-tuned agent pipeline could address this consensus failure mode.

Additionally, validating and extending the trust-aware orchestration approach across varied datasets and domains beyond image classification to include video analysis, language tasks, and multimodal reasoning will be important for demonstrating generalizability and broader impact.

Ultimately, embedding self-monitoring and self-improvement mechanisms within agents may enable autonomous adaptation and increased system resilience, paving the way for more robust and scalable Agentic AI architectures applicable to complex real-world problems.

V. CONCLUSION

This study introduced a trust-aware Agentic AI framework that integrates modular vision-language agents, confidence calibration metrics, and retrieval-augmented reasoning to achieve reliable visual classification without task-specific fine-tuning. Through three progressive experimental configurations, we demonstrated that structured trust orchestration can bridge the performance gap between zero-shot and fine-tuned models, achieving 85.63% accuracy compared to a 48.13% baseline and approaching the 97.50% accuracy of supervised adaptation. The system's hierarchical architecture, combining independent agent perception with meta-cognitive arbitration and CLIP-based image retrieval, enables interpretable, scalable, and trustworthy AI decision-making suitable for deployment in high-stakes domains requiring transparency and accountability.

STATEMENT ON AI WRITING ASSISTANCE

ChatGPT and Grammarly were utilized to enhance grammatical accuracy and refine sentence structure; all AI-generated revisions were thoroughly reviewed and edited for relevance.

APPENDIX

SUPPLEMENTARY FIGURES AND TABLES

See Figure 9,
See Figure 10,
See Figure 11,
Table 2,
Table 3,
Table 4,
Table 5,
Table 6,
Table 7.

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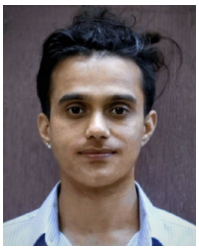

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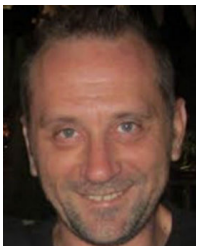

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